

Questions

SetA

Carefully read the question and design the **Assignment** class with appropriate variables and methods such that the following main method produces the expected output. For simplicity, assume that only simulation and coding assignment types are supported here.

Note that, when a specific practice time is not given, a simulation task time reduces by one fifth and a coding task reduces to one half of its original time.

Tester Code	Expected Output
<pre>public class AssignmentTester { public static void main(String[] args) { System.out.println(Assignment.status()); System.out.println("1 >====="); Assignment a1 = new Assignment("CSE111", "Simulation", 3); Assignment a2 = new Assignment("MAT120", "Coding", 4.8); System.out.println(a2.details()); System.out.println(a1.details()); System.out.println("2 >====="); System.out.println(Assignment.status()); System.out.println("3 >====="); a1.submit(); a2.practice(0.25); Assignment a3 = new Assignment("PHY112", "Simulation", 5); System.out.println(Assignment.status()); System.out.println("4 >====="); System.out.println(a3.details()); System.out.println(a2.details()); System.out.println(a1.details()); System.out.println("5 >====="); a3.practice(); a2.practice(); Assignment a4 = new Assignment("CSE220", "Coding", 3.2); System.out.println("6 >====="); System.out.println(a4.details()); System.out.println(Assignment.status()); } }</pre>	<pre>Pending Assignments: 0 Simulation and 0 Coding tasks Total time: 0.0 hours 1 >===== Assignment ID: C-02 Coding: MAT120 [4.8 Hr] Assignment ID: S-01 Simulation: CSE111 [3.0 Hr] 2 >===== Pending Assignments: 1 Simulation and 1 Coding tasks Total time: 7.8 hours 3 >===== S-01 assignment submitted. C-02 time reduced to 3.6 Hr Pending Assignments: 1 Simulation and 1 Coding tasks Total time: 8.6 hours 4 >===== Assignment ID: S-03 Simulation: PHY112 [5.0 Hr] Assignment ID: C-02 Coding: MAT120 [3.6 Hr] Assignment ID: S-01 Simulation: Submitted [0.0 Hr] 5 >===== S-03 time reduced to 4.0 Hr C-02 time reduced to 1.8 Hr 6 >===== Assignment ID: C-04 Coding: CSE220 [3.2 Hr] Pending Assignments: 1 Simulation and 2 Coding tasks Total time: 9.0 hours</pre>

SetB

Carefully read the question and design the **Subscription** class with appropriate variables and methods such that the following main method produces the expected output. For simplicity, assume that only Streaming and Storage services are supported here. Note that, when a specific cashback rate is not given, 50% cashback is applied to a storage service and 20% cashback is applied for a streaming service.

Tester Code	Expected Output
<pre>public class SubscriptionTester { public static void main(String[] args) { System.out.println(Subscription.status()); System.out.println("1 >====="); Subscription c1 = new Subscription("Netflix", "Streaming", 3000); Subscription c2 = new Subscription("Drive", "Storage", 4000); System.out.println(c2.details()); System.out.println(c1.details()); System.out.println("2 >====="); System.out.println(Subscription.status()); System.out.println("3 >====="); c1.unSubscribe(); c2.cashBack(0.25); Subscription c3 = new Subscription("Prime", "Streaming", 3000); System.out.println(Subscription.status()); System.out.println("4 >====="); System.out.println(c3.details()); System.out.println(c2.details()); System.out.println(c1.details()); System.out.println("5 >====="); c3.cashBack(); c2.cashBack(); Subscription s4 = new Subscription("Dropbox", "Storage", 5000); System.out.println("6 >====="); System.out.println(s4.details()); System.out.println(Subscription.status()); } }</pre>	<pre>Total Subscriptions: 0 Streaming and 0 Storage services Total fee: 0.0 Tk 1 >===== Subscription ID: S-02 Storage: Drive => 4000.0 Tk Subscription ID: S-01 Streaming: Netflix => 3000.0 Tk 2 >===== Total Subscriptions: 1 Streaming and 1 Storage services Total fee: 7000.0 Tk 3 >===== S-01 Subscription has been cancelled. S-02 fees reduced to 3000.0 Tk Total Subscriptions: 1 Streaming and 1 Storage services Total fee: 6000.0 Tk 4 >===== Subscription ID: S-03 Streaming: Prime => 3000.0 Tk Subscription ID: S-02 Storage: Drive => 3000.0 Tk Subscription ID: S-01 Streaming: Cancelled => 0.0 Tk 5 >===== S-03 fees reduced to 2400.0 Tk S-02 fees reduced to 1500.0 Tk 6 >===== Subscription ID: S-04 Storage: Dropbox => 5000.0 Tk Total Subscriptions: 1 Streaming and 2 Storage services Total fee: 8900.0 Tk</pre>

LabExam 1 Solutions

[Set A solution]

```
class Assignment { // /// 1
    static int total_count = 0;
    static double total_time = 0;
    static int sim = 0;
    static int code = 0;
    String assignment;
    String assign_no;
    String assign_type;
    double assign_hour;

    Assignment(String assign, String type, double time) { // /// 1.5
        total_count += 1; // 0.5
        total_time += time;
        if (type.equals("Simulation")) { // 0.5
            sim += 1;
        } else {
            code += 1;
        }
        this.assignment = assign; // 0.5
        this.assign_type = type;
        this.assign_hour = time;
        this.assign_no = type.charAt(0) + "-0" + total_count;
    }

    void submit() { // /// 1.5
        total_time -= this.assign_hour; // 0.5
        System.out.println(assign_no+" assignment submitted."); // 0.5
        this.assign_hour = 0;
        this.assignment = "Submitted";
        if (assign_type.equals("Simulation")) { // 0.5
            sim -= 1;
        } else if (assign_type.equals("Coding")) {
            code -= 1;
        }
    }

    //////////////////////////////////////
    static String status() { // /// 1.5
        String out = "Pending Assignments:\n";
        out += sim + " Simulation and " + code + " Coding tasks\n";
        out += "Total time: " + total_time + " hours";
        return out;
    }

    String details() { // /// 1.5
        String out = "Assignment ID: " + assign_no + "\n";
        out += assign_type + ": " + assignment + " [" + assign_hour + " Hr]";
        return out;
    }

    void practice() { // /// 1.5
```

```
// 0.5
if (assign_type.equals("Simulation")) {
    practice(0.2); // 1
} else if (assign_type.equals("Coding")) {
    practice(0.5);
}
}

void practice(double ratio) { // 1.5
    // 1
    total_time -= assign_hour;
    ratio = 1 - ratio;
    assign_hour = ((int)(100 * assign_hour * ratio)) / 100.0;
    total_time += assign_hour;
    // 0.5
    System.out.println(assign_no+" time reduced to " + assign_hour + " Hr");
}
}
```

[Set B solution]

```
class Subscription { // /// 1
    static int total_count = 0;
    static double total_fee = 0;
    static int stream = 0;
    static int storage = 0;
    String subscription;
    String sub_id;
    String sub_type;
    double sub_fee;

    Subscription(String service, String type, double fee) { // /// 1.5
        total_count += 1; // 0.5
        total_fee += fee;
        if (type.equals("Streaming")) { // 0.5
            stream += 1;
        } else {
            storage += 1;
        }
        this.subscription = service; // 0.5
        this.sub_type = type;
        this.sub_fee = fee;
        this.sub_id = type.charAt(0) + "-0" + total_count;
    }

    void unSubscribe() { // /// 1.5
        total_fee -= this.sub_fee; // 0.5
        System.out.println(sub_id+" Subscription has been cancelled."); // 0.5
        this.sub_fee = 0;
        this.subscription = "Cancelled";
        if (sub_type.equals("Streaming")) { // 0.5
            stream -= 1;
        } else if (sub_type.equals("Storage")) {
            storage -= 1;
        }
    }

    ///////////////////////////////////////////////////
    static String status() { // /// 1.5
        String out = "Total Subscriptions:\n";
        out += stream + " Streaming and " + storage + " Storage services\n";
        out += "Total fee: " + total_fee + " Tk";
        return out;
    }

    String details() { // /// 1.5
        String out = "Subscription ID: " + sub_id + "\n";
        out += sub_type + ": " + subscription + " => " + sub_fee + " Tk";
        return out;
    }

    void cashBack() { // /// 1.5
        // 0.5
        if (sub_type.equals("Streaming")) {
```

```
        cashBack(0.2); // 1
    } else if (sub_type.equals("Storage")) {
        cashBack(0.5);
    }
}

void cashBack(double ratio) { // 1.5
    // 1
    total_fee -= sub_fee;
    ratio = 1 - ratio;
    sub_fee = ((int)(100 * sub_fee * ratio)) / 100.0;
    total_fee += sub_fee;
    // 0.5
    System.out.println(sub_id+" fees reduced to " + sub_fee + " Tk");
}
}
```

Marking Rubric

For both sets:

Criteria	Points
Basic Class Design → [1] All class related syntax.	1
Constructor: Assignment(Str, Str, double) / Subscription(Str, Str, double) → Distribution / Breakdown in solution	1.5
Method: submit():void / unSubscribe():void → Distribution / Breakdown in solution	1.5
Method: practice(double):void / cashBack(double):void → Distribution / Breakdown in solution	1.5
Overloaded Method: practice():void / cashBack():void → Distribution / Breakdown in solution	1.5
static Method: status():String → Distribution / Breakdown in solution	1.5
Method: details():String → Distribution / Breakdown in solution	1.5
Total:	10